



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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SATHYAMANGALAM - 638 401 ERODE DISTRICT TAMIL NADU INDIA

Ph: 04295-226000 / 221289 Fax: 04295-226666 E-mail: stayahead@bitsathy.ac.in Web: www.bitsathy.ac.in

22AI503/22AM503 MACHINE LEARNING

UNIT II – DATA PREPROCESSING

Data Visualization and Measuring Data Similarity & Dissimilarity

1. Introduction

1.1 Definition

Data visualization is the graphical representation of data and information to help humans perceive patterns, trends, clusters, and anomalies that are difficult to detect from raw numbers alone. Measuring data similarity and dissimilarity quantifies how alike or different two data objects are — a core operation underpinning clustering, classification, and information retrieval algorithms.

Definitions:

Data Visualization: Graphical encoding of data to reveal structure, patterns, and outliers.

Similarity: A measure (0 to 1) of how alike two data objects are. Higher = more similar.

Dissimilarity (Distance): A measure (≥ 0) of how different two objects are. Lower = more similar.

1.2 Importance in Machine Learning

Similarity and dissimilarity measures are the mathematical foundation of kNN classifiers, k-Means clustering, hierarchical clustering, DBSCAN, anomaly detection, recommendation systems, and information retrieval. Choosing the wrong metric can completely change clustering outcomes or kNN classification results.

1.3 Role in Data Preprocessing Pipeline

Visualization is used in EDA to understand data distributions, attribute relationships, and cluster structures before preprocessing. Similarity measures drive distance-based preprocessing steps such as imputation by nearest neighbours and outlier detection using distance-to-centroid.

2. Learning Outcomes

Understand

- The purpose and classification of data visualization types
- Minkowski distance family (Manhattan, Euclidean, Chebyshev)
- Similarity measures for binary, categorical, and text data

Apply

- Select the appropriate visualization for a given data/question combination
- Compute Euclidean, Manhattan, and Cosine distances manually
- Build similarity/dissimilarity matrices for kNN and clustering

Analyze

- Compare distance metrics and justify selection based on attribute types and use case
- Interpret t-SNE and PCA scatter plots for cluster analysis
- Evaluate the curse of dimensionality impact on distance metrics

3. Detailed Technical Concepts

3.1 Data Visualization Types and Selection Guide

The choice of visualization depends on the nature of the data (attribute types) and the analytical question (distribution, comparison, relationship, composition, or high-dimensional exploration).

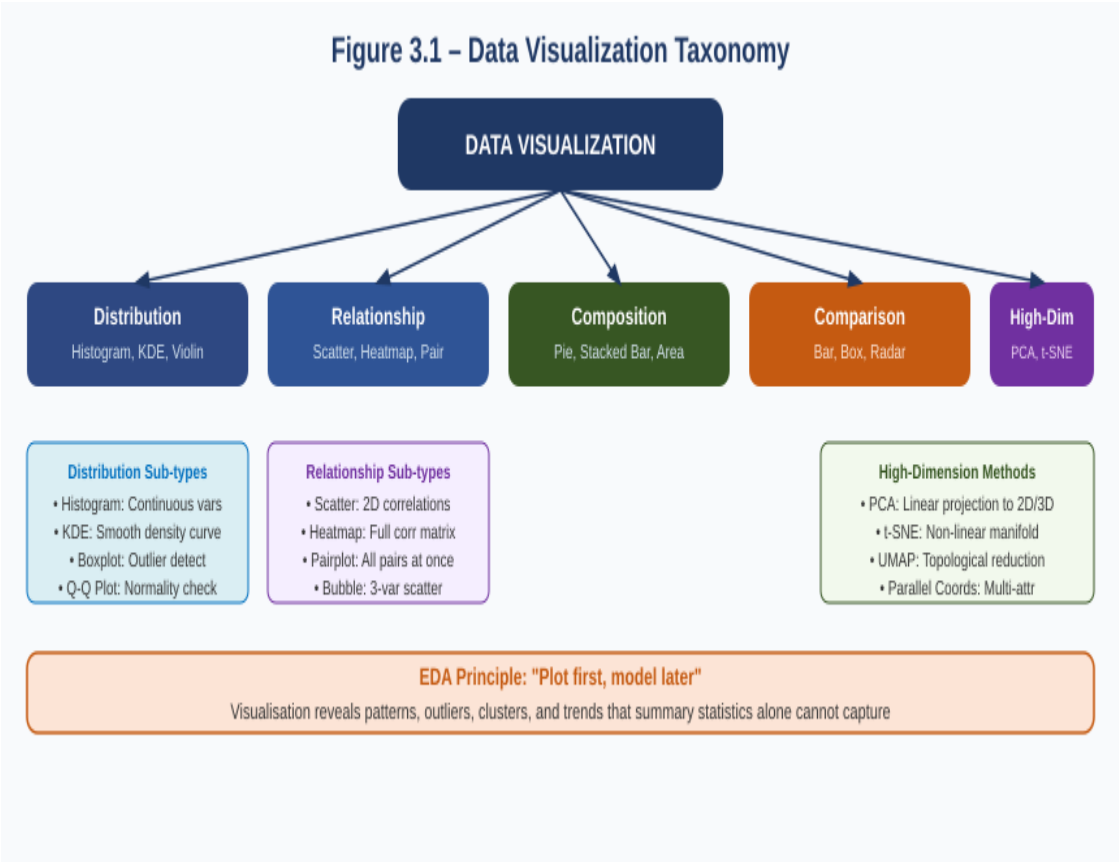


Figure 3.1 – Data Visualization Taxonomy by Purpose

3.2 Core Visualization Techniques

Visualization	Best For	Attribute Type	Key Insight Revealed
Histogram	Frequency distribution of one attribute	Numeric (continuous)	Skewness, modality, spread
Kernel Density (KDE)	Smooth probability density curve	Numeric (continuous)	Distribution shape without bin bias
Boxplot	Distribution + outlier detection	Numeric	5-number summary, outlier fences

Scatter Plot	Relationship between 2 numeric attributes	2 Numeric	Correlation, clusters, linearity
Bar Chart	Category frequencies or comparisons	Nominal / Ordinal	Class imbalance, category dominance
Heatmap (Corr)	Correlation structure across all pairs	All numeric attrs	Multicollinearity, feature groupings
Pairplot	All pairwise scatter + diagonals	Multiple numeric attrs	Quick multi-attribute overview
Q-Q Plot	Normality testing	Numeric	Points on diagonal = normal
t-SNE / UMAP	High-dimensional cluster visualisation	Any (after encoding)	Natural clusters in 2D projection
Parallel Coords	Multi-attribute comparison per instance	Multiple numeric attrs	Cluster trajectories across dimensions
Violin Plot	Distribution shape per group	Numeric grouped by categ	Modality within each category

3.3 High-Dimensional Visualization

3.3.1 Principal Component Analysis (PCA) for Visualization

PCA projects high-dimensional data onto the top 2 or 3 principal components (directions of maximum variance) for scatter plot visualization. It preserves global structure but may distort local cluster boundaries.

Steps: (1) Standardize features, (2) Compute covariance matrix, (3) Find eigenvectors (PCs), (4) Project data onto PC1–PC2, (5) Plot scatter coloured by class.

3.3.2 t-SNE (t-Distributed Stochastic Neighbor Embedding)

t-SNE converts pairwise similarities between data points into probability distributions and minimises KL divergence between high-dim and low-dim distributions. Preserves local neighborhood structure — excellent for revealing clusters but distances between clusters are not meaningful.

Key hyperparameter: Perplexity (5–50) — controls the effective number of neighbours. Run multiple times with different random seeds to verify cluster stability.

3.4 Measuring Data Similarity and Dissimilarity

For n data objects, similarity/dissimilarity is stored in an $n \times n$ matrix. Dissimilarity matrices satisfy: $d(x,x)=0$, $d(x,y)=d(y,x)$ (symmetry), and $d(x,z) \leq d(x,y)+d(y,z)$ (triangle inequality).

3.5 Distance Metrics for Numeric Attributes

3.5.1 Minkowski Distance (General Form)

$d(x,y) = (\sum_i |x_i - y_i|^p)^{1/p}$ where $p \geq 1$ is the order parameter.

- $p = 1$: Manhattan Distance (L1 Norm): $d = \sum |x_i - y_i|$. Robust to outliers. Used in LASSO regularisation.
- $p = 2$: Euclidean Distance (L2 Norm): $d = \sqrt{\sum (x_i - y_i)^2}$. Most common. Default in kNN and k-Means.

- $p \rightarrow \infty$: Chebyshev Distance (L^∞ Norm): $d = \max|x_i - y_i|$. Used in warehouse path planning and chess king moves.

3.5.2 Mahalanobis Distance

$d(x,y) = \sqrt{[(x-y)^T S^{-1} (x-y)]}$ where S is the covariance matrix. Accounts for attribute scale and correlation. Scale-invariant and correlation-aware. Used in anomaly detection and QDA.

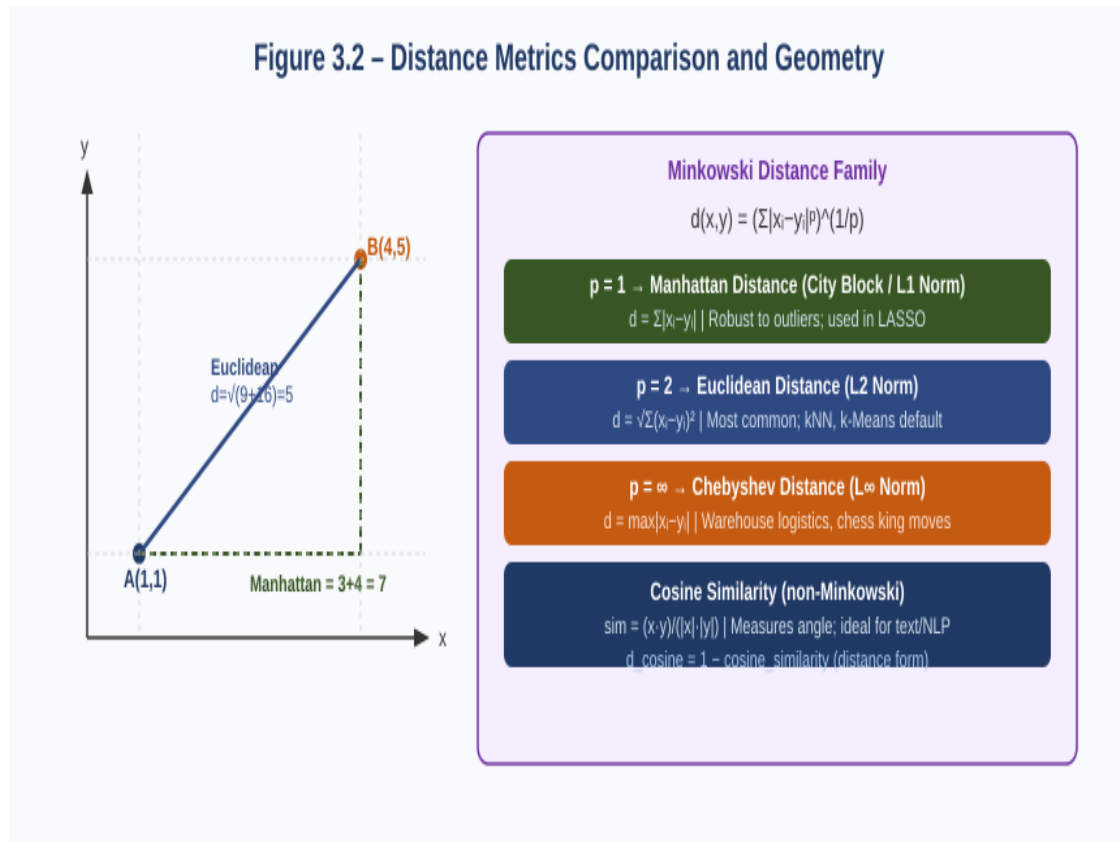


Figure 3.2 – Distance Metrics Comparison: Manhattan, Euclidean, Chebyshev, Cosine

3.6 Similarity Measures

3.6.1 Cosine Similarity (for Text / Sparse Vectors)

$\text{sim}(x,y) = (x \cdot y) / (||x|| \times ||y||)$ where $\cdot$ is dot product and $||x||$ is L_2 norm.

Range: $[-1, 1]$ for general vectors; $[0, 1]$ for TF-IDF/bag-of-words (non-negative). Measures the angle between vectors, not their magnitude. Essential for NLP: two documents with same word frequencies but different lengths are equally similar.

Cosine Distance: $d = 1 - \text{cos_sim}(x,y)$. Used in text clustering, recommendation engines.

3.6.2 Jaccard Similarity (for Sets / Binary Attributes)

$$J(A,B) = |A \cap B| / |A \cup B| = f_{11} / (f_{10} + f_{01} + f_{11})$$

Suitable for asymmetric binary attributes and set comparisons. Used in: market basket analysis (items co-purchased), document deduplication (shingling), social network similarity.

3.6.3 Simple Matching Coefficient (SMC) (Symmetric Binary)

$$\text{SMC}(x,y) = (f_{11} + f_{00}) / (f_{11} + f_{10} + f_{01} + f_{00})$$

Used when both 0 and 1 are equally meaningful (e.g., gender). Note: Jaccard ignores f_{00} (both negative) making it preferable for sparse binary data.

3.6.4 Correlation-Based Similarity

Pearson: $\text{sim}(x,y) = [\sum (x_i - \bar{x})(y_i - \bar{y})] / [n \times \sigma_x \times \sigma_y]$. Range: $[-1, 1]$. Negative similarity possible (inverse relationships). Used in collaborative filtering recommendation systems.

3.7 Similarity Measures for Mixed Data

Gower Distance handles mixed types in a single formula:

$$d_Gower(x,y) = (1/d) \times \sum_i w_i \times d_i(x_i,y_i)$$

where d_i is the appropriate metric for attribute type i : $|x_i - y_i|/\text{range}$ for numeric, $1 - J$ for binary, $1 - I(x_i = y_i)$ for nominal. Used in R's `daisy()` function and mixed-type clustering.

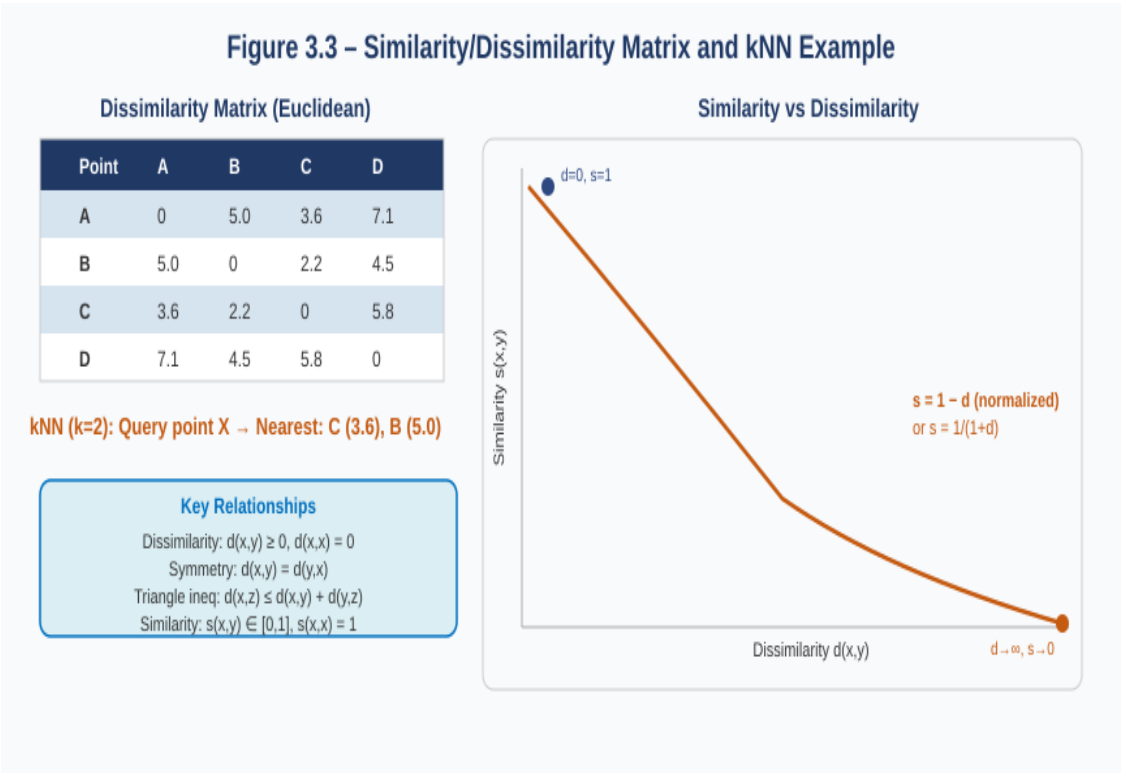


Figure 3.3 – Similarity/Dissimilarity Matrix and kNN Example

3.8 Curse of Dimensionality

As dimensionality d increases, Euclidean distances lose discriminative power. All pairwise distances converge to approximately the same value, making nearest neighbour searches meaningless. For $d > 20-30$, feature selection or dimensionality reduction (PCA) is essential before distance-based computation.

4. Architecture — Choosing the Right Metric

Data Type	Recommended Metric	Algorithm Use Case	Notes
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Numeric, same scale	Euclidean (L2)	kNN, k-Means, SVM	Standardize first if scales differ
Numeric, different scale	Manhattan (L1)	LASSO, robust kNN	Less sensitive to large deviations
Text / Sparse vectors	Cosine Similarity	TF-IDF clustering, IR	Magnitude-independent measure
Binary (asymmetric)	Jaccard Coefficient	Market basket, doc dedup	Ignores double-negative matches
Binary (symmetric)	Simple Matching Coeff	Medical symptoms, surveys	Both 0 and 1 are equally important
Mixed types	Gower Distance	Mixed-attribute clustering	Handles nominal+ordinal+numeric
High-dim, correlated	Mahalanobis Distance	Anomaly detection, QDA	Accounts for covariance structure
Ordered categories	Spearman Rank Corr	Ordinal rating similarity	Non-parametric, rank-based

5. Industry Tools and Technologies

5.1 Commercial Visualization Tools

Tool	Best Feature	Similarity/Distance Support
Tableau	Interactive drag-drop dashboards	Via R/Python integration
Power BI	DAX + Python/R visuals	Custom distance metrics in Python
Qlik Sense	Associative analytics model	Semantic similarity in extensions
Plotly Dash	Interactive web dashboards from Python	Full scipy.spatial.distance support

5.2 Open-Source Tools

- matplotlib/seaborn: hist(), boxplot(), scatter(), heatmap(), pairplot()
- Plotly: Interactive 3D scatter, parallel coordinates, animated EDA
- scikit-learn: euclidean_distances(), cosine_similarity(), pairwise_distances()
- scipy.spatial.distance: cdist() supporting 30+ metrics
- sklearn.manifold.TSNE, umap-learn: High-dimensional visualization
- Bokeh, Altair: Grammar-of-graphics visualization in Python

6. Real-World Applications

Domain	Visualization Used	Similarity Metric	Application
E-Commerce	Heatmap of product co-purchases	Jaccard similarity	Collaborative filtering recommendations
NLP	Word cloud, t-SNE of embeddings	Cosine similarity	Document clustering, semantic search

Genomics	Dendrogram, heatmap (gene expr)	Euclidean / Correlation	Gene expression cluster analysis
Medical Imaging	PCA 2D scatter by diagnosis	Mahalanobis distance	Anomaly detection in patient data
Finance	Correlation heatmap of stocks	Pearson correlation	Portfolio diversification analysis

7. Advantages and Limitations

Advantages	Limitations
Visualization reveals hidden patterns invisible in tables	High-dimensional data ($d > 3$) cannot be directly visualised
Distance metrics enable all unsupervised learning algorithms	Euclidean distance fails on high-cardinality categorical vars
t-SNE reveals natural cluster structure without labels	t-SNE is non-deterministic and computationally expensive $O(n^2)$
Gower distance handles real-world mixed-type datasets	Computing $n \times n$ distance matrix requires $O(n^2)$ memory for large n
Cosine similarity is scale-invariant for text analysis	Jaccard ignores feature magnitudes — not suitable for numeric data

8. Case Study

CASE STUDY

Case: Netflix Movie Recommendation via Cosine Similarity

Problem: Netflix (2006 Netflix Prize competition) needed to predict user ratings for unseen movies. Initial approach using Euclidean distance on raw ratings performed poorly (RMSE = 1.05).

Dataset: 100 million ratings, 480,189 users, 17,770 movies.

Investigation: User-movie rating vectors are extremely sparse (most users rate $< 1\%$ of movies). Euclidean distance penalizes missing ratings as zeros, making active and inactive users appear dissimilar regardless of taste.

Solution: Switched to Cosine Similarity on mean-centered rating vectors. This approach: (1) centers ratings per user (removing user bias), (2) computes cosine angle between rating vectors ignoring zero-rated items.

Visualization: t-SNE of user rating embeddings revealed clear genre-preference clusters (Action fans, Romance fans, Documentary fans) that were invisible in raw Euclidean space.

Findings: Cosine similarity with SVD matrix factorization achieved RMSE = 0.88 — beating Netflix's own algorithm by 10%, winning the \$1M Netflix Prize.

Conclusion: Choosing the right similarity metric for sparse, user-generated data dramatically improved recommendation quality. Cosine similarity is the standard for collaborative filtering.

9. Scenario-Based Learning

SCENARIO 1

SCENARIO 1: Wrong Distance Metric in Customer Segmentation

Background: A retail analytics team applied k-Means clustering with default Euclidean distance to segment customers using {age, income, ZIP code, category preference}.

Technical Environment: scikit-learn KMeans, pandas, 200,000 customer records.

Problem Statement: k-Means produced clusters dominated by ZIP code (e.g., 600001 vs 600100 = 99 unit difference) while age and income had smaller ranges, making geography overshadow all other features.

Available Evidence: `df.describe()` shows ZIP range = 99, Age range = 60, Income range = 90,000. ZIP dominates Euclidean distance by a factor of 1,500x vs Age.

Investigation: Feature scaling was missing. Euclidean distance on unstandardised features is dominated by the attribute with the largest numeric range.

Analysis: Standardize all numeric features (StandardScaler: mean=0, std=1). Encode ZIP as nominal (not numeric). Use Gower distance for the mixed-type dataset.

Findings: After standardization and proper encoding, silhouette score improved from 0.18 to 0.61.

Clusters became meaningful (Young urban, Suburban family, Senior rural).

Recommended Solution: Always StandardScale numeric features before computing Euclidean distance.

For nominal attributes, use one-hot encoding or Gower distance.

SCENARIO 2

SCENARIO 2: t-SNE Reveals Hidden Sub-clusters in Medical Data

Background: A cancer diagnosis model showed 78% accuracy on a 50-feature gene expression dataset. The data scientist suspected hidden subtypes of cancer that the single-class label missed.

Technical Environment: Python, sklearn TSNE, UMAP, K-Means, 500 patient samples × 50 gene features.

Problem Statement: PCA scatter plot (PC1 vs PC2) showed a single overlapping blob — no visible cluster separation. Team assumed data was homogeneous.

Available Evidence: Explained variance ratio: PC1 = 18%, PC2 = 12% (total 30% — PCA discards 70% of variance, losing cluster structure).

Investigation: Apply t-SNE with perplexity=30 on standardized data. UMAP applied for comparison.

Analysis: t-SNE scatter revealed 4 distinct clusters corresponding to Luminal A, Luminal B, HER2+, and Basal subtypes of breast cancer.

Findings: Stratifying training data by t-SNE-identified subtype and training 4 separate models improved average accuracy from 78% to 91%. Subtype-specific biomarker genes were identified from each cluster.

Recommended Solution: Use PCA first (fast), then t-SNE/UMAP for detailed cluster exploration. Always try multiple perplexity values (10, 30, 50). Validate t-SNE clusters with domain knowledge.

SCENARIO 3

SCENARIO 3: Similarity Matrix for Plagiarism Detection

Background: A university's academic integrity system needs to detect plagiarised assignment submissions. 500 student Python scripts need pairwise comparison.

Technical Environment: Python, TF-IDF vectorizer, cosine similarity, `scipy.spatial`, pandas.

Problem Statement: Simple file comparison (exact match) misses paraphrased code. Token-level similarity needed.

Available Evidence: 500 scripts, average 200 lines each. 12 suspected plagiarism pairs identified manually by instructor.

Investigation: Convert each script to TF-IDF vector (code token frequencies). Compute 500×500 cosine similarity matrix. Flag pairs with `sim > 0.85`.

Analysis: Cosine similarity captures similar function names and code structure even if variable names are changed. Heatmap of similarity matrix reveals plagiarism clusters as bright diagonal blocks.

Findings: System flagged 18 suspicious pairs (cosine sim > 0.85). Manual review confirmed 15 true

plagiarism cases. 3 false positives (same algorithm taught in class).
 Recommended Solution: Combine cosine similarity with Jaccard of AST (Abstract Syntax Tree) node sets for more robust code similarity. Threshold at 0.85 with manual review above 0.90.

10. Common Errors and Troubleshooting

Error	Cause	Impact	Fix
Euclidean on unstandardised features	Missing StandardScaler	High-range features dominate distance	Apply StandardScaler before computing distances
Cosine on dense numeric data	Confusing text and numeric metrics	Ignores magnitude; inappropriate	Use Euclidean for dense numeric vectors
Jaccard on numeric (not binary) data	Misapplying set similarity	Incorrect intersection computation	Use Jaccard only for binary/set data
t-SNE distances as cluster centroids	Misinterpreting t-SNE	Distances between clusters meaningless	Use t-SNE only for visual cluster detection, not distance computation
n×n matrix for n>100K records	Memory-inefficient distance comp	Out of Memory error (100K ² =10B floats)	Use approximate NN (FAISS, Annoy) for large n

11. Key Terminologies

Term	Definition
Similarity	A measure in [0,1] of how alike two data objects are; $s(x,x)=1$
Dissimilarity	A metric $d(x,y) \geq 0$ of how different two objects are; $d(x,x)=0$
Euclidean Distance	L2 norm: $\sqrt{\sum (x_i - y_i)^2}$. Default for numeric data.
Manhattan Distance	L1 norm: $\sum x_i - y_i $. Robust, used in LASSO regularisation.
Minkowski Distance	Generalised: $(\sum x_i - y_i ^p)^{1/p}$. Includes L1, L2, L^∞ .
Cosine Similarity	Angle-based similarity = $(x \cdot y) / (x \cdot y)$. Ideal for text.
Jaccard Coefficient	Set similarity = $ A \cap B / A \cup B $. For binary/sparse data.
Gower Distance	Mixed-type distance combining numeric, ordinal, and nominal metrics.
Mahalanobis Distance	Scale-invariant, covariance-aware distance using inverse covariance matrix.
Dissimilarity Matrix	n×n matrix storing pairwise distances between all data objects.
t-SNE	t-Distributed Stochastic Neighbor Embedding for 2D/3D cluster visualization.
PCA	Principal Component Analysis: linear projection onto max-variance directions.
Curse of Dimensionality	Phenomenon where distances lose discriminative power as d increases.

12. Topic Summary

UNIT II – Topic 3

DATA VISUALIZATION AND SIMILARITY/DISSIMILARITY — SUMMARY

- Visualization Types: Distribution (histogram, KDE, box), Relationship (scatter, heatmap), Composition (pie, stacked), Comparison (bar, violin), High-Dim (PCA, t-SNE).
- Minkowski Family: $p=1$ (Manhattan/ L_1), $p=2$ (Euclidean/ L_2), $p=\infty$ (Chebyshev/ L_∞).
- Cosine Similarity: angle-based, scale-invariant, standard for text/NLP and sparse vectors.
- Jaccard: for binary/set data (asymmetric). SMC: for symmetric binary attributes.
- Gower Distance: unified metric for mixed numeric, ordinal, and nominal data.
- Always StandardScale numeric features before Euclidean distance computation.
- t-SNE reveals local cluster structure; PCA preserves global structure. Use both for EDA.
- Curse of dimensionality: distance-based methods degrade above ~30 dimensions; use PCA first.
- Tools: seaborn, matplotlib, Plotly, scikit-learn (euclidean_distances, cosine_similarity), scipy, UMAP, Tableau.